# Krisna Joko Purjianto

Software Engineer

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Surakarta, Central Java, Indonesia (Open to relocate)

## SUMMARY

Software Engineer with 2+ years of experience building and maintaining production-grade web applications, REST APIs, and transaction-driven systems using PHP/Laravel and Golang. Experienced in backend development, third-party integrations, MySQL query optimization, authentication, service migration, and CI/CD, with hands-on ownership from development through production. Comfortable contributing across the stack with JavaScript, Vue.js, React.js, and Next.js, collaborating with cross-functional teams, and working independently on technical problems. Strong interest in building reliable systems, improving performance, and taking ownership of system architecture.

## EXPERIENCE

**Software Engineer (Full-time) |** *Nov 2024 – Present*

*PT Transformasi Data Digital*

- Maintained and developed client websites focused on top-up systems and digital transaction flows using PHP (Native OOP and Laravel) and MySQL..
- Monitored ad bot performance and ensured smooth client transaction pipelines.
- Participated in API development for new client feature integrations and maintained API documentation as requirements and implementations evolved.
- Integrated third-party APIs (Coda, domain provisioning) into client top-up websites as new feature requirements arose.
- implementing end to end atomic database operations and high-precision decimal calculations to prevent data anomalies and ensure absolute financial accuracy.
- Collaborated with cross-functional teams to keep communication clear and aligned, establishing structured discussion habits that reduced miscommunication across the team.

**Software Engineer (Side Project) |** *August 2025 – Present*

*TokTik Indonesia*

- Built and maintained the backend using Laravel, Golang, MySQL, and REST APIs for an electricity token and digital top-up platform serving 2,301+ customers (still increasing).
- Driven by cost efficiency, load-tested and migrated high-traffic transaction endpoints to Go after exhausting Laravel optimizations, including query tuning, caching, and OPcache. Successfully increased throughput from ~50 req/s to ~200 req/s, maximizing the 2-core/2GB server capacity at ~95% utilization without requiring costly vertical scaling.
- Implemented clean architecture with repository pattern, service layer,and raw SQL for the Go service, plus token-based authentication and authorization middleware for the Go service, connected the Laravel and Go services via gRPC for callbacks, and documented service architecture and API interactions for system maintenance.
- Handled asynchronous payment and provider callbacks across payment gateway and digital product aggregator, including transaction states where the expected second callback was delayed or not received.
- Cached product data and authenticated user profiles in Redis as serialized JSON strings with TTL-based expiry to reduce MySQL reads.
- Implemented idempotent callback handling using transaction order IDs and status validation to prevent duplicate payment and top-up processing.
- Integrated payment gateway and digital product aggregator provider, implementing idempotent callback handling using transaction order IDs to safely process asynchronous events, delayed callbacks, and prevent duplicate payments.
- Implemented automated unit testing in the CI/CD pipeline, running tests before Docker image builds and automated deployment (Alibaba Cloud).
- Built the Agent portal of TokTik with Next.js and integrated it with the Go backend APIs, covering withdrawal and redemption flows.

**Software Engineer (Freelance / Project-based) |** *Mar 2025 – Aug 2025*

*PT UMBI Teknologi Indonesia*

- Led team for maintenance, optimization, and development website event Solo Raya Great Sale using laravel and mysql as the database.

- Responsible for data management, database optimization, and infrastructure stability monitoring.
- Optimized database performance across 23,000+ merchant records, 5 million+ transaction records, and 212,041 users through indexing and query optimization.
- Planned and delegated tasks efficiently to team members based on capacity and skill.
- Coordinated with admin team to handle user and tenant issue reports rapidly.
- Drove collaborative team culture, streamlining delegation and resulting in faster turnaround on issue reports.

**Software Engineer (Freelance / Project-based) |** *Apr 2024 – Jul 2024*

*PT UMBI Teknologi Indonesia*

- Maintained and developed the Solo Raya Great Sale event website.
- Updated UI, fixed bugs and errors, and ensured consistent website performance.
- Added new features and adjusted existing ones based on event requirements.

## SKILLS

**Programming Languages:** PHP, Golang, JavaScript, TypeScript, Node.js, HTML5, SQL

**Frameworks & Libraries:** Laravel, Gin, React.js, Next.js, Vue.js

**Databases:** MySQL, PostgreSQL,Redis

**DevOps & Infrastructure:** Linux, Git, Github, GitHub Actions (CI/CD), Docker, Server Deployment,gRPC

**Other:** Microservices Architecture, REST API, Payment Gateways, Unit Testing, Alibaba Cloud, Agile/Scrum, SDLC

## EDUCATION

**Bachelor of Information Technology** GPA: 3.74

University Duta Bangsa Surakarta | Sep 2020 – Sep 2024

First-authored academic publication in LOFIAN Vol. 4 No. 1 (August 2024): IoT-based household energy monitoring system using ESP32 and PZEM-004T sensors.

## LANGUAGES

- Indonesia (Native)
- English (A2)

## CERTIFICATIONS

**Certificate of Excellence - Cybersecurity Engineer Bootcamp | Merdeka Siber Batch #21 (February 2026)**

- Hands-on practice in vulnerability assessment and penetration testing: reconnaissance, exploitation, and reporting.
- Practiced against common web vulnerabilities: SQL Injection, XSS, CSRF, Broken Access Control, IDOR, Unrestricted File Upload, OWASP Top 10.
- Applied security principles in a production, building systems that are functional and secure.

## PROJECTS

**Portal Nevalis — ERP System**

*Laravel, ReactJS, MySQL, Ajax, jQuery, REST API, GitHub Actions*

- Led end-to-end architecture and development of an ERP system as team lead under PT Transformasi Data Digital, driving technical decisions through team discussions and taking ownership of final decisions when needed, from system design to client communication with SDLC Development method and conducting code reviews to maintain code quality and consistency.
- Implemented SSE for asynchronous background import processing, preventing controller timeout on large data imports.
- Leveraged IndexedDB for client-side draft auto-saving on complex multi-field forms, syncing to server on submission.
- Applied ReactJS selectively on one high-complexity page to manage JavaScript state without overengineering.
- Set up GitHub Actions CI/CD pipeline on shared hosting via SSH for automated deployment.

**Protowa — WhatsApp API Platform**

*React.js, Laravel, Node.js, PostgreSQL, Microservice*

- Designed and built a 3-stack microservice architecture (React frontend, Laravel backend, Node.js WhatsApp service) as sole developer.
- Migrated from whatsapp-web.js (Puppeteer, ~40MB/session) to Baileys (WebSocket-based) for significantly lower resource footprint.

- Solved Baileys credential persistence through iterative approach: moved from impractical file storage to in-memory caching, then to a final solution persisting both ephemeral keys and session data to PostgreSQL, ensuring full session survival across restarts.
- Implemented SSE combined with Node.js Event Emitter for real-time QR scan notifications outside the HTTP request cycle.

**Incubator System**

*Golang, PostgreSQL, Linux, ESP32 (Arduino/C++), Vue.js, TypeScript, Github Action*

- Architected a multi-device IoT backend using Golang to handle high-frequency concurrent requests from multiple sensor devices sending data every second.
- Built Golang backend with PostgreSQL and PgBouncer connection pooling, load tested with k6 achieving 200 req/s on a 4-core 4GB RAM server at ~50% CPU usage.
- Implemented real-time monitoring using Server-Sent Events (SSE) with Vue.js and TypeScript dashboard, enabling live sensor data display.
- Integrated ESP32 microcontrollers communicating directly with the Golang REST API via HTTPS, sending sensor readings every second.

**atsin.id — ATS-Friendly CV Generator**

*Vue.js, TypeScript, Node.js, Playwright, IndexedDB*

- Built a privacy-first CV generator using IndexedDB for client-side storage no login required, personal data never leaves the browser.
- Implemented PDF generation via Node.js and Playwright for single-click download experience.
- Built with Vue.js and TypeScript as a deliberate production learning ground for both technologies.